Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/17/2007
PI Name:	Paloski, William H Ph.D.		
Project Title:	Spatial Reorientation of Sensorimotor Balance Control in Altered Gravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:	PI moved to University of Houston in September 2008, from NASA JSC.		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	10/01/1999	**End Date:**	03/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	Shuttle		
Flight Assignment:	NOTE: Project extended until 03/31/2008 per PI (5/07) Pre-/Postflight Only In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Wood, Scott Ph.D. (Universities Space Research Association) Kaufman, Galen Ph.D. (University of Texas Medical Branch) Reschke, Millard Ph.D. (NASA Johnson Space Center) Black, Franklin M.D. (Legacy Health Sysytem)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Disruption of balance control is one of the most significant post-flight manifestations of sensorimotor adaptation to microgravity. This investigation examines changes in spatial processing of sensorimotor function following adaptation to microgravity, and explores the feasibility of forcing changes between motor program sets optimized for different gravitoinertial environments. This study involves pre- and post-flight measurements on eleven astronaut subjects (first-time fliers) following short-duration Shuttle missions (11-13 days). Each astronaut participates in one familiarization training session (including standard posturography testing), three pre-flight data sessions, and four post-flight data sessions on R+0, 2, 3, and 7/8 days. Each test session includes posture measurements, and three sessions (L-60, L-30 and R+3) include additional posture measurements immediately after exposure to short-radius centrifugation. In addition, testing on normative subjects (ongoing) support data interpretation and establish ground-based controls. Our first specific aim is to examine adaptive changes in the spatial reference frame used for coding orientation and motion as a function of space flight. By examining the effects of head tilt on balance control, we will test our hypothesis that there is a reorientation of central vestibular processing from a gravitational frame of reference to a head frame of reference as a function of adaptation to microgravity. Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch centrifugation. This study specifically addresses Neurovestibular Adaptation questions of the Critical Path Roadmap (CPR Risks 15 and 16) related to the physiological basis for disruption of balance control and spatial disorientation following g-transitions. Additional post-flight posturography measures obtained during this study will expand our existing database and increase our ability to predict recovery curves from mathematical models for individuals based on pre-flight postural performance. The incorporation of head tilts during posturography addresses our general hypothesis for the physiological basis for the disruption of balance control following g-transitions, and provides an enhanced technique to determine crew readiness to return to normal activities following return to Earth. The incorporation of short-radius centrifugation during the post-flight period addresses the ability of crewmembers to adapt to multiple gravitoinertial environments and switch between them, e.g., as required for intermittent exposure to artificial gravity. The results of this study will also assess the potential risk for disrupting the re-adaptation process by exposure to discordant sensory stimuli during low levels of centrifugation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has demonstrated the increased sensitivity of computerized dynamic posturography (CDP) with the incorporation of static and dynamic head movements and a sway-referenced platform. This enhanced CDP protocol seems more suitable for monitoring astronaut fitness for return to duty, and may be useful in the early detection of balance disorders.
Task Progress:	During this past fiscal year, astronaut data collection was completed on the final 2 subjects on STS-116 in December 2006. The ground-based control study was also resumed. This study will utilize normative subjects that are gender, age and height matched to their astronaut counterparts and undergo the same test schedule. This control study will allow us to evaluate learning effects using our enhanced balance control protocol independent of the effects of space-flight. Preliminary findings: During pre-flight testing all subjects adequately compensated for different head orientations with respect to gravity during eyes-closed, fixed-support conditions (SOT-2). Consistent with the results from our previous normative study (Paloski et al., Gait & Posture 23: 315-23, 2006), postural stability was decreased during dynamic head tilts with eyes-closed and sway-referenced support (SOT-5). Postural sway occurred primarily in the anterior-posterior (AP) direction versus medial-lateral (ML). Postural stability was decreased on R+0 for all conditions with absent vision and altered proprioceptive feedback (SOT-5). The performance decrements on trials requiring dynamic head movements were most striking: all eleven subjects fell on at least one SOT-5 “Dynamic” trial on R+0, suggesting that this stability threatening condition should be avoided by crewmembers during normal daily activities. It is perhaps equally important to note how the same head tilt conditions using a fixed support (SOT-2) failed to detected the sensorimotor change, especially since this is the typical paradigm used clinically (Romberg test). Performance recovery for head erect conditions followed similar trajectories to those previously reported by our laboratory. Static and dynamic head tilt trials had similar recovery time constants as head erect, but at lower performance levels. Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch axis centrifugation. There were no significant differences between pre-SAC and post-SAC measures. There were also no significant differences between L-30 and R+3 sessions in motion perception reports and eye movements during centrifugation. Conclusions: The highly disrupted balance control performance during head tilts after flight is consistent with our hypothesis that changes in the central nervous system processing of otolith information contributes to the disruption of balance control following g-transitions. Adding head movements to SOT 5 enhances the sensitivity of the balance control measures for astronauts, perhaps making it more functionally relevant test for this highly competitive subject group. Absence of similar findings for the static tilt conditions suggest that processing of dynamic canal-otolith afferent information is severely disrupted early after flight, but that this can be compensated for by limiting head movements. In addition, the absence of similar findings during dynamic head tilts on a fixed support surface indicates the use of somatosensory input to compensate for adaptive changes in vestibular function. Based on the results of this study, head tilts during posturography have been implemented as a medical requirement for functional neurological assessment following short- and long-duration space missions. The standard Romberg test would underestimate the deficits observed on the computerized sway-referenced platform. While post-flight posture performance decrements have been reported following exposures to short-radius off-axis pitch rotation, limited exposure to similar pitch centrifugation profiles on R+3 (nominal return to duty day) did not substantially disrupt postural performance in this study. Further work will be required to identify the incidence and mechanisms responsible for the paroxysmal losses of spatial orientation following space flight known as “flashbacks.”
Bibliography Type:	Description: (Last Updated: 02/09/2021)

Abstracts for Journals and Proceedings	Paloski WH. "Sensitive quantitative assessment of balance disorders. " Aerospace Medical Association 78th Annual Scientific Meeting, New Orleans LA, May 13-17, 2007. Aviat Space Environ Med. 2007 Mar;78(3):230. , Mar-2007
Abstracts for Journals and Proceedings	Paloski, WH, Black FO, Hwang EY, Reschke MF, Wood SJ. "Spatial reorientation of balance control following short-duration space flight." 16th IAA Humans in Space Symposium, Beijing, China, May 2007. 16th IAA Humans in Space Symposium, Beijing, China, May 2007. , May-2007
Abstracts for Journals and Proceedings	Paloski, WH, Black FO, Kaufman GD, Reschke MF, Wood SJ. "Spatial reorientation of sensorimotor balance control in altered gravity." NASA Human Research Program Investigators' Workshop, League City TX, February 2007. NASA Human Research Program Investigators' Workshop, League City TX, February 2007. , Feb-2007